Rebuttal Tables

Table 1: Top Four CABS Variants on the Open LLM Leaderboard 2 (Sparsity=0.75, Official Benchmark Results)

Models	**IFEval**	**BBH**	**MATH**	**GPQA**	**MUSR**	**MMLU**	**AVG**
Tsunami-0.5-7b	74.00	36.14	50.45	7.83	12.21	37.92	36.43
fq2.5-7b	73.99	**36.36**	46.22	6.94	17.54	37.92	36.50
CABS-v0.1(Ours)	75.06	35.84	47.96	8.50	14.17	37.84	36.56
CABS-v0.2(Ours)	74.18	36.28	49.02	7.61	14.86	37.75	36.61
CABS-v0.3(Ours)	75.70	35.96	49.32	7.61	15.24	37.80	36.94
CABS-v0.4(Ours)	**75.83**	**36.36**	48.49	7.72	15.17	37.73	36.88

Table 2: Evaluation of Merged Mistral Models: Instruction-Tuned vs. Math-Tuned Variants (Sparsity=0.75, numbers in parentheses indicate improvement over Task Arithmetic).

Method	λ	**IFEval**	**GSM8K**	**AVG**
Mistral-7b-v0.1	–	31.18	36.85	34.02
Mistral-7b-v0.1-instruct	–	49.88	35.03	42.46
math-sft	–	23.38	70.89	47.14
Ideal Model	–	49.88	70.89	60.39
Task Arithmetic	0.5	39.81	63.99	51.90
TIES-Merging	0.8	**41.13**	62.40	51.77(-0.13)
DARE	2.0	38.61	61.41	50.01(-1.89)
CABS (Ours)	0.8	41.01	**64.37**	**52.69(+0.79)**

Table 3: Performance of different merging methods on DTD and EuroSAT tasks using ViT-B-32 models.(Sparsity=0.90, numbers in parentheses indicate improvement over Task Arithmetic)

Method	λ	**DTD**	**EuroSAT**	**AVG**
Base	–	43.19	32.93	38.06
Ideal Model	–	78.88	98.85	88.87
Task Arithmetic	0.68	68.51	98.15	83.33
TIES-Merging	1.79	69.31	97.96	83.64(+0.31)
DARE	6.10	68.19	98.04	83.11(-0.22)
CABS (Euro-first)	1.96	**72.50**	**98.56**	**85.53(+2.20)**
CABS (DTD-first)	1.90	72.13	**98.56**	85.34(+2.01)

Table 4: Expanded version of Table 8 in the full paper, with additional results for the Localize-and-Stitch method. Other results are consistent with Table 8.

Model	RTE	MRPC	AVG
Base Model	47.29	68.38	57.84
Ideal Model	79.42	91.18	85.30
Task Arithmetic	73.29	87.01	80.15
Dataless L&S (Top-5%)	47.29	68.63	57.96(-22.19)
Dataless L&S (Top-10%)	51.62	71.57	61.60(-18.55)
Consensus TA	73.29	87.01	80.15(+0.00)
CABS (Ours)	**74.01**	**88.97**	**81.49(+1.34)**

Table 5: Expanded version of Table 1 in the full paper, with additional results for the Localize-and-Stitch method. Other results are consistent with Table 1.

Model	RTE	MRPC	CoLA	SST-2	AVG
Base Model	47.29	68.38	69.13	50.92	58.93
Ideal Model	79.42	91.18	85.04	94.04	87.42
Task Arithmetic	76.32	90.83	69.68	81.37	79.55
Dataless L&S (Top-5%)	47.29	69.36	69.32	63.30	62.32(-17.23)
Dataless L&S (Top-10%)	58.84	74.51	73.06	85.44	72.96(-6.59)
CABS (Ours)	**76.89**	**92.09**	**74.73**	**83.09**	**81.70(+2.25)**

Table 6: Expanded version of Table 9 in the full paper, with additional results for the Localize-and-Stitch method. Other results are consistent with Table 9.

Model	RTE	MRPC	CoLA	SST-2	RACE	SQuAD	AVG
Base Model	47.29	68.38	69.13	50.92	27.97	34.53	49.70
Ideal Model	79.42	91.18	85.04	94.04	71.71	79.82	83.54
Task Arithmetic	67.15	79.41	72.00	85.78	56.21	38.82	66.56
Dataless L&S (Top-5%)	47.29	68.87	69.32	56.88	50.52	11.74	50.77(-15.79)
Dataless L&S (Top-10%)	62.82	76.26	71.52	75.34	55.43	26.19	61.26(-5.30)
CABS (mrscsqra)(Ours)	**71.12**	**82.35**	**74.30**	90.14	57.38	41.56	69.47(+2.91)
CABS (csrmrasq)(Ours)	68.95	82.11	73.92	**90.83**	**58.97**	**42.96**	**69.62(+3.06)**

Table 7: Evaluation of 7-task model merging on GPT-2. All results except Base Model, DARE and CABS are from FusionBench[1].

Method	CoLA	MNLI	MRPC	QNLI	QQP	RTE	SST-2	**AVG**
Base Model	69.1	33.0	68.4	49.2	63.2	52.7	50.1	55.09
Ideal Model	76.8	82.1	80.4	88.3	89.6	65.3	91.2	81.96
Task Arithmetic	68.7	68.6	69.6	70.5	81.8	47.3	83.6	70.01
TIES-Merging	68.4	**71.4**	68.4	69.6	**82.4**	47.7	81.8	69.96(-0.05)
DARE	**69.2**	64.7	**69.9**	**73.0**	79.3	47.3	83.3	69.53(-0.48)
CABS (Ours)	68.5	68.5	68.1	70.4	81.9	**49.5**	**85.7**	**70.37(+0.36)**

Table 8: Average parameter magnitude of task vectors across different tasks (original values $\times 10^3$ for readability).

Task Vectors	RTE	MRPC	CoLA	SST-2	QNLI	QQP	MNLI
Magnitude ($\times 10^3$)	0.710	0.851	1.383	3.379	3.872	6.828	7.300

Table 9: Comparison of sparsity strategies (sparsity=0.9).

Method	**RTE**	**MRPC**	**AVG**
Fine-tuned on RTE	79.42	25.98	52.70
Fine-tuned on MRPC	47.29	91.18	69.24
Task Arithmetic	73.29	87.01	80.15
+ DARE	72.92	**88.24**	80.58(+0.43)
+ Magnitude (layer-wise)	**74.73**	86.03	80.38 (+0.23)
+ Magnitude (row-wise)	74.06	87.05	80.61 (+0.46)
+ SparseGPT	72.92	87.75	80.34 (+0.19)
+ WANDA	73.29	87.50	80.40 (+0.25)
BS (Ours)	74.37	88.23	**81.30 (+1.08)**

References

[1] Anke Tang, Li Shen, Yong Luo, Han Hu, Bo Du, and Dacheng Tao. Fusionbench: A comprehensive benchmark of deep model fusion. *arXiv preprint arXiv:2406.03280*, 2024.